

TCC70xx MCU BSP

API Specification for Pulse Density Modulation

Rev. 0.10 [A]

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Preliminary version

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1 INTRODUCTION

This document describes the pulse density modulation device driver (hereinafter referred to as PDM driver) of TCC70xx MCU BSP. For more detailed information on pulse density modulation in the TCC70xx MCU Sub-system, refer to Chapter 12 Pulse Density Modulation (PDM) in "*TCC70xx Full Specification*"[1].

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

PDM	Pulse Density Modulation
PMM	Pulse Monitoring Module

3 SOURCE FILES

3.1 Location of Source Files

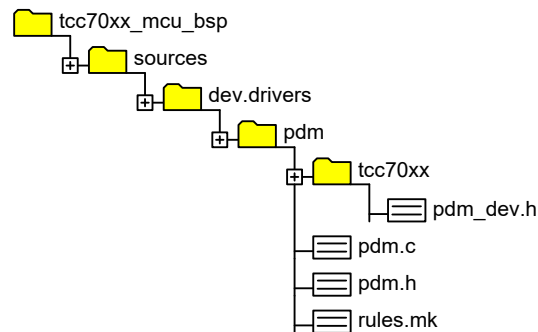


Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- pdm_dev.h
 - PDM specific header
- pdm.c
 - PDM common source code
- pdm.h
 - PDM common header including API & Defines
- rules.mk
 - Rules used for build

4 CONSTANTS

4.1 Connection Type

Constant values for the PDM output connection type to the PMM. The default value for PDM connection type is EXTERNAL. The PDM connection type is used only to check safety mechanism.

Declaration

```
#define PDM_CONNECT_INTERNAL (0x00000000UL)
#define PDM_CONNECT_EXTERNAL (0x00000001UL)
```

Description

Value	Description
PDM_CONNECT_INTERNAL	PDM monitoring module checks the PDM output that is internally connected. (internal loopback)
PDM_CONNECT_EXTERNAL	PDM monitoring module checks the PDM output that is externally connected. (external loopback)

Remark

None

4.2 Channel Information Values

Constant values for the operating information of PDM channel

Declaration

```
#define PDM_SAFETY_FEATURE_ENABLED

#define PDM_STATE_BUSY          (1UL)
#define PDM_STATE_IDLE         (0UL)

#define PDM_ON                  (1UL)
#define PDM_OFF                 (0UL)

#ifdef PDM_SAFETY_FEATURE_ENABLED
#define PMM_ON                  (1UL)
#define PMM_OFF                 (0UL)
#endif

#define PDM_TOTAL_MODULES       (3UL) /* The PWM controller consist of 3 PWM modules (PWM-0/1/2). */
#define PDM_TOTAL_CH_PER_MODULE (4UL) /* Each PWM module has four separate PWM blocks (PWM-A/B/C/D). */
#define PDM_TOTAL_CHANNELS      PDM_TOTAL_MODULES * PDM_TOTAL_CH_PER_MODULE /* total 12channels */
```

Description

Value	Description
<i>PDM_SAFETY_FEATURE_ENABLED</i>	<i>Define safety feature</i>
<i>PDM_STATE_BUSY</i>	<i>Busy state of PDM</i>
<i>PDM_STATE_IDLE</i>	<i>Idle state of PDM</i>
<i>PDM_ON</i>	<i>PDM is enabled.</i>
<i>PDM_OFF</i>	<i>PDM is disabled.</i>
<i>PMM_ON</i>	<i>PMM is enabled.</i>
<i>PMM_OFF</i>	<i>PMM is disabled.</i>
<i>PDM_TOTAL_MODULES</i>	<i>The number of PDM modules</i>
<i>PDM_TOTAL_CH_PER_MODULE</i>	<i>The number of PDM channels per module</i>
<i>PDM_TOTAL_CHANNELS</i>	<i>The number of total PDM channels</i>

Remark

None

4.3 PDM Mode Value

Constant values for the PDM mode

Declaration

```
#define PDM_PERI_CLOCK          125UL * 1000UL * 1000UL /* PDM MAX clk 125Mhz */

#define PDM_DIVID_MAX          3UL /* Clock divide max value 3 (divide 16) */

/*Operation Mode Phase 1 & 2*/
#define PDM_OUTPUT_MODE_PHASE_1 0x0001UL
#define PDM_OUTPUT_MODE_PHASE_2 0x0009UL

/*Operation Mode Register 1 & 2 & 3*/
#define PDM_OUTPUT_MODE_REGISTER_1 0x0002UL
#define PDM_OUTPUT_MODE_REGISTER_2 0x0004UL
#define PDM_OUTPUT_MODE_REGISTER_3 0x0006UL

#define REG_OUTPUT_PATTERN_HIGH 0xFFFFFFFFUL
#define REG_OUTPUT_PATTERN_LOW 0x00000000UL
```

Description

Value	Description
PDM_PERI_CLOCK	PDM peripheral clock value
PDM_DIVID_MAX	Maximum clock divide value, divide by 16
PDM_OUTPUT_MODE_PHASE_1	Phase mode 1
PDM_OUTPUT_MODE_PHASE_2	Phase mode 2
PDM_OUTPUT_MODE_REGISTER_1	Register mode 1
PDM_OUTPUT_MODE_REGISTER_2	Register mode 2
PDM_OUTPUT_MODE_REGISTER_3	Register mode 3
REG_OUTPUT_PATTERN_HIGH	High output pattern of register mode
REG_OUTPUT_PATTERN_LOW	Low output pattern of register mode

Remark

Refer to Chapter 12 for each output mode in "*TCC70xx Full Specification*" [1].

5 STRUCTURES

5.1 PDMModeConfig_t

Structure to store the PDM output mode information of each channel

Declaration

```
typedef struct _PDMModeConfig_
{
    uint32 mcPortNumber;
    uint32 mcOperationMode;
    uint32 mcClockDivide;
    uint32 mcLoopCount;
    uint32 mcInversedSignal;
    uint32 mcPeriodNanoSec1;
    uint32 mcDutyNanoSec1;
    uint32 mcPeriodNanoSec2;
    uint32 mcDutyNanoSec2;
    uint32 mcPosition1;
    uint32 mcPosition2;
    uint32 mcPosition3;
    uint32 mcPosition4;
    uint32 mcOutPattern1;
    uint32 mcOutPattern2;
    uint32 mcOutPattern3;
    uint32 mcOutPattern4;
    uint32 mcMaxCount;
} PDMModeConfig_t;
```

Description

Value	Description
mcPortNumber	PDM output port number. Refer to “TCC70xx Chip Specification”.
mcOperationMode	PDM operation Mode, Phase Mode 1, 2 and Register Mode 1, 2, 3
mcClockDivide	Clock divide value ■ 0: Divide by 2 ■ 1: Divide by 4 ■ 2: Divide by 8 ■ 3: Divide by 16
mcLoopCount	Period cycle count. If this value is 0, PDM operates infinite loop mode.
mcInversedSignal	Inversed output signal
mcPeriodNanoSec1	Period1 value in Nano second scale for phase mode
mcDutyNanoSec1	Duty1 value in Nano second scale for phase mode
mcPeriodNanoSec2	Period2 value in Nano second scale for phase mode

Value	Description
<i>mcDutyNanoSec2</i>	<i>Duty2 value in Nano second scale for phase mode</i>
<i>mcPosition1</i>	<i>Register PSTN1 value for Phase mode 1 and 2</i>
<i>mcPosition2</i>	<i>Register PSTN2 value for Phase mode 1 and 2</i>
<i>mcPosition3</i>	<i>Register PSTN3 value for Phase mode 2</i>
<i>mcPosition4</i>	<i>Register PSTN4 value for Phase mode 2</i>
<i>mcOutPattern1</i>	<i>Register OUT_PTN1 value for Register mode 1, 2, and 3</i>
<i>mcOutPattern2</i>	<i>Register OUT_PTN2 value for Register mode 2 and 3</i>
<i>mcOutPattern3</i>	<i>Register OUT_PTN3 value for Register mode 2 and 3</i>
<i>mcOutPattern4</i>	<i>Register OUT_PTN4 value for Register mode 2 and 3</i>
<i>mcMaxCount</i>	<i>Register MAX_CNT value for Register mode 3. The count range is from 1 to 127.</i>

Remark

None

5.2 PDMChannelHandler_t

Structure to store the PDM channel handle information on each channel

Declaration

```
typedef struct _PDMChannelHandler_
{
    uint32_t      chModuleId;
    uint32_t      chChannelId;
    uint32_t      chEnable;
    uint32_t      chOutSignalInIdle;
    uint32_t      chIdleState;
    PDMModeConfig_t chModeCfgInfo;

#ifdef PDM_SAFETY_FEATURE_ENABLED
    uint32_t      chPMMEnable;
    uint32_t      chPMMInputCnct;
    uint32_t      chPMMFaultStatus;
    uint32_t      chPMMTimeoutValue;
    uint32_t      chPMMDutyMargin;
    uint32_t      chPMMErrChannel;
#endif
} PDMChannelHandler_t;
```

Description

Value	Description
<i>chModuleId</i>	PDM Module ID PDM 0/1/2
<i>chChannelId</i>	PDM Channel ID 0/1/2/3
<i>chEnable</i>	PDM Channel Status, PDM_ON or PDM_OFF
<i>chOutSignalInIdle</i>	PDM output signal in idle mode
<i>chIdleState</i>	Idle state of PDM
<i>chModeCfgInfo</i>	Configuration value of PDM operation mode
<i>chPMMEnable</i>	PMM Monitoring Status, PMM_ON or PMM_OFF.
<i>chPMMInputCnct</i>	Input connection type of monitoring module, Internal or External connection
<i>chPMMFaultStatus</i>	Fault status detected by PMM
<i>chPMMTimeoutValue</i>	PMM timeout value
<i>chPMMDutyMargin</i>	PMM duty margin value
<i>chPMMErrChannel</i>	PMM error channel. If this value is TRUE, it means the PDM channel fault occurs.

Remark

None

6 APIs

6.1 PDM_EnableClock

Enables peripheral clock for all PDM modules. The default clock rate is set to 125 MHz (Maximum is 125 MHz) and should be modified if necessary.

Declaration

```
void PDM_EnableClock  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6.2 PDM_DisableClock

Disables peripheral clock for all PDM modules.

Declaration

```
void PDM_DisableClock  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6.3 PDM_Init

Initializes all PDM channels. If this function is called, [PDMChannelHandler](#) structure for all PDM channels is filled with default value. And the PDM peripheral clock rate for each module is set and the PDM peripheral clock is enabled. If the PDM safety feature is enabled, this function registers FMU interrupt handler and enables FMU interrupts.

Declaration

```
sint32 PDM_Init  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6.4 PDM_Deinit

De-initializes all PDM channels. If this function is called, [PDMChannelHandler](#) structure for all channels is filled with 0.

Declaration

```
void PDM_Deinit  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6.5 PDM_GetChannelStauts

Gets the PDM channel status (BUSY or IDLE). When the PDM channel is enabled or the PDM output mode is set, this function is called.

Declaration

```
uint32 PDM_GetChannelStauts  
(  
    uint32 uiChannel  
);
```

Parameters

Value	Description
<i>uiChannel</i>	<i>[in] PDM Channel value 0 to 11</i>

Return

Value	Description
<i>PDM_STATE_BUSY</i>	<i>PDM is in Busy state. (= 0)</i>
<i>PDM_STATE_IDLE</i>	<i>PDM is in IDLE state. (= 1)</i>

Remark

None

6.6 PDM_Enable

Starts generating the PDM signal. For more information, refer to Chapter 12 in “*TCC70xx Full Specification*”. (PDM Enable Register (OP_EN)) [1]. PDM EN sequence is as follows: Set EN → Set VUP → Set TRIG.

Declaration

```
SALRetCode_t PDM_Enable
(
    uint32 uiChannel,
    uint32 uiMonitoring
);
```

Parameters

Value	Description
<i>uiChannel</i>	<i>[in] PDM Channel value 0 to 11</i>
<i>uiMonitoring</i>	<i>[in] Enable the PMM monitoring module for safety feature.</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

6.7 PDM_Disable

Stops generating the PDM output signal.

Declaration

```
SALRetCode_t PDM_Disable  
(  
    uint32 uiChannel,  
    uint32 uiMonitoring  
);
```

Parameters

Value	Description
<i>uiChannel</i>	<i>[in] PDM Channel value 0 to 11</i>
<i>uiMonitoring</i>	<i>[in] Disable the PMM monitoring module for safety feature.</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

6.8 PDM_SetConfig

Sets configuration values of the PDM registers and. Sets *PSTNx* or *OUT_PTNx* register according to the PDM operation mode.

Declaration

```
SALRetCode_t PDM_SetConfig  
(  
    uint32          uiChannel,  
    PDMModeConfig_t * pModeConfig  
);
```

Parameters

Value	Description
<i>uiChannel</i>	<i>[in]</i> PDM Channel value 0 to 11
<i>pModeConfig</i>	<i>[in]</i> Pointer of configuration structure. Refer to PDMModeConfig_t .

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

7 HOW TO USE PDM DRIVER

The PDM driver can generate pulse signal according to user configuration such as the PDM mode, duty, period, etc. To get more information on signal configuration of the PDM mode, refer to the description of PDM signal in "*TCC70xx Full Specification*". [1]

7.1 Operation of PDM

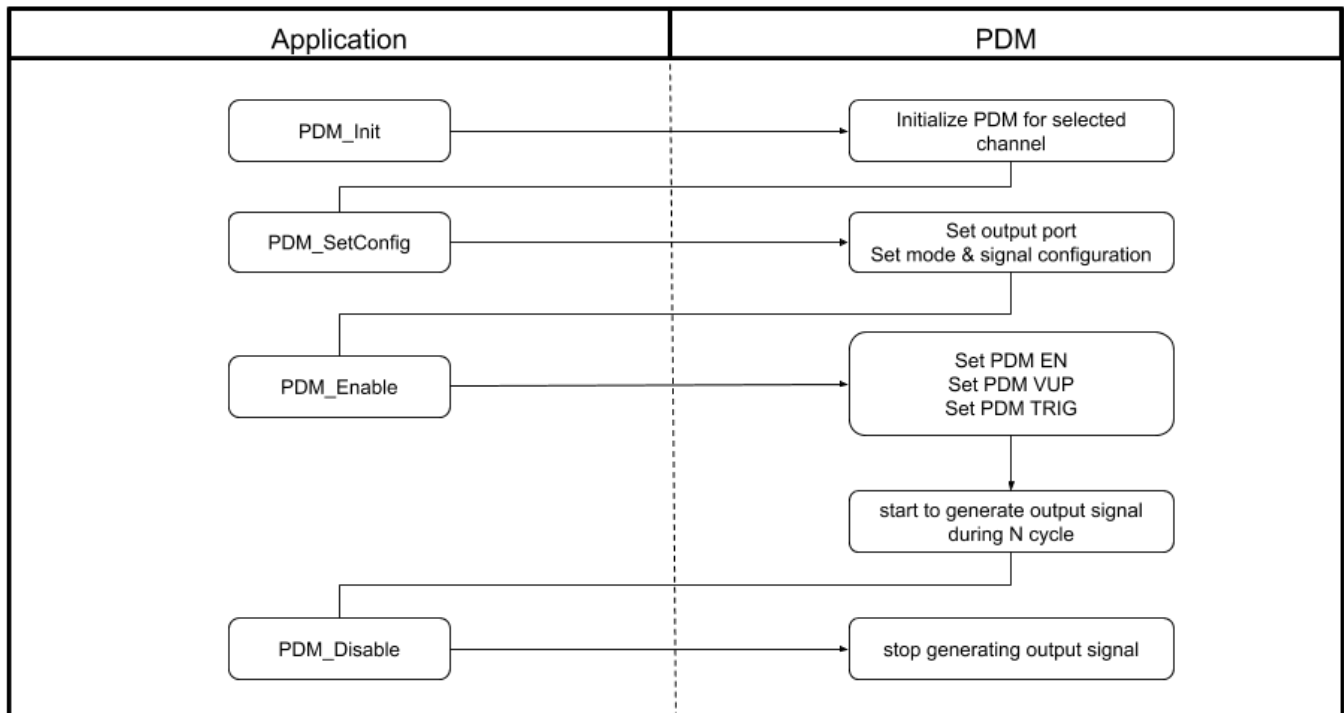


Figure 7.1 Diagram for PDM Driver

7.2 Example: Using PDM Driver

```

{
    // channel initializes
    PDM_Init();

    // set configuration. (port, mode, signal)
    PDM_SetConfig();

    // start to generate
    PDM_Enable();
}
  
```

8 REFERENCES

- [1] TCC70xx Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

9 REVISION HISTORY

Rev. 0.10: 2021-09-03

- Preliminary version release

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